Figure 1: Injection objects — max-1 CPP triples on the slack simplex

Profile $c = (2, 1, 1)$, $d = 4$. *Slack* $s = (s_1, s_2, s_3)$, $s_1{+}s_2{+}s_3 = 4$.

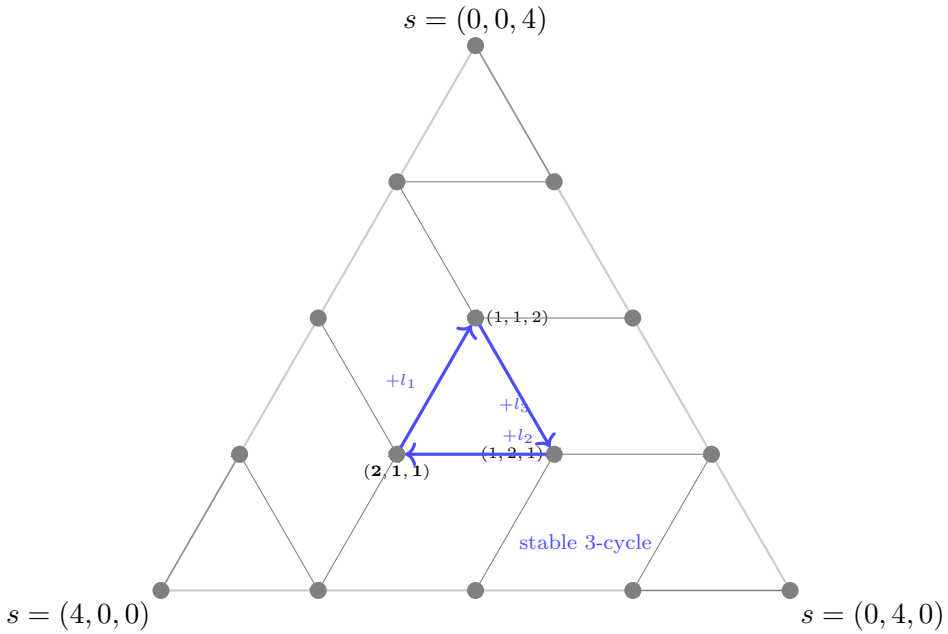
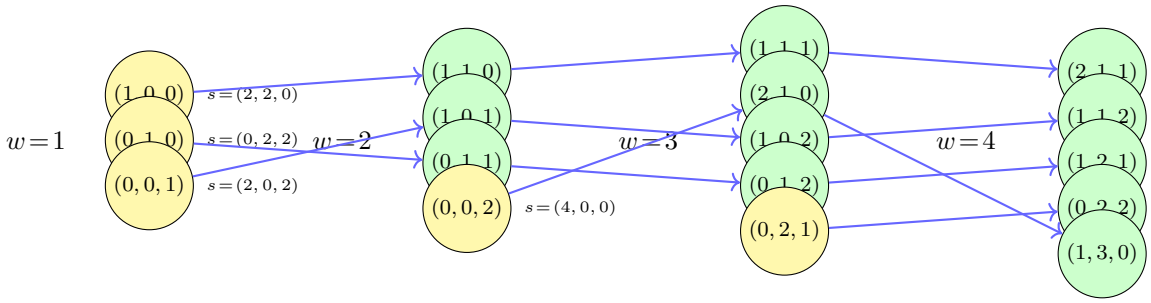


Figure 2: The injection ϕ weight by weight

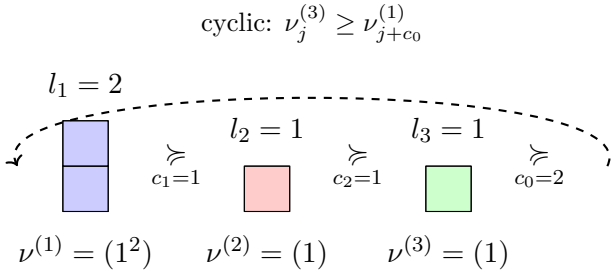
$c = (2, 1, 1)$. *Each triple* (l_1, l_2, l_3) *shown with its slack. Yellow = fixed point (not in image of* ϕ*).*



Fixed points give $h_1(q)$: w=1: 3, w=2: 1, w=3: 1, $w\geq 4$: 0 $\implies$ $h_1 = 3q + q^2 + q^3$

Figure 3: A max-1 CPP triple as column partitions

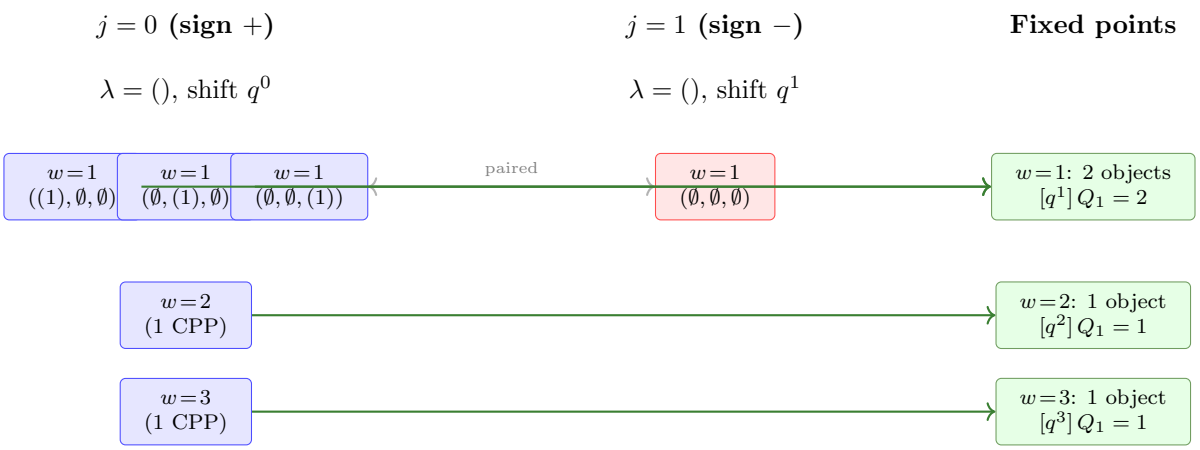
$c = (2, 1, 1)$, *triple* $(l_1, l_2, l_3) = (2, 1, 1)$, *weight* $w = 4$, *slack* $s = (2, 1, 1)$.



Slacks: $s_1 = l_3 + c_0 - l_1 = 1 + 2 - 2 = 1$, $s_2 = l_1 + c_1 - l_2 = 2 + 1 - 1 = 2$, $s_3 = l_2 + c_2 - l_3 = 1 + 1 - 1 = 1$

Figure 4: Signed objects for the involution ($n = 1$, $c = (2, 1, 1)$)

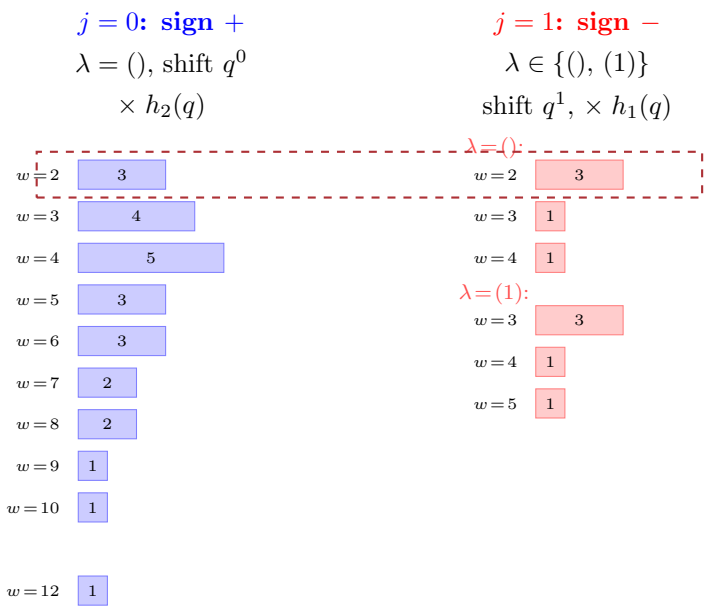
$$Q_1 = h_1 - q \cdot h_0 = (3q + q^2 + q^3) - q = 2q + q^2 + q^3$$



Summary: 5 objects total (4 positive from h_1, 1 negative from $-q \cdot h_0$). The involution pairs 1 positive with 1 negative at weight 1. Fixed points: $2 + 1 + 1 = 4 = Q_1(1)$.

Figure 5: Signed objects for the involution ($n = 2$, $c = (2, 1, 1)$)

$$Q_2 = h_2 - q(1+q)\, h_1 + q^3 \cdot h_0$$



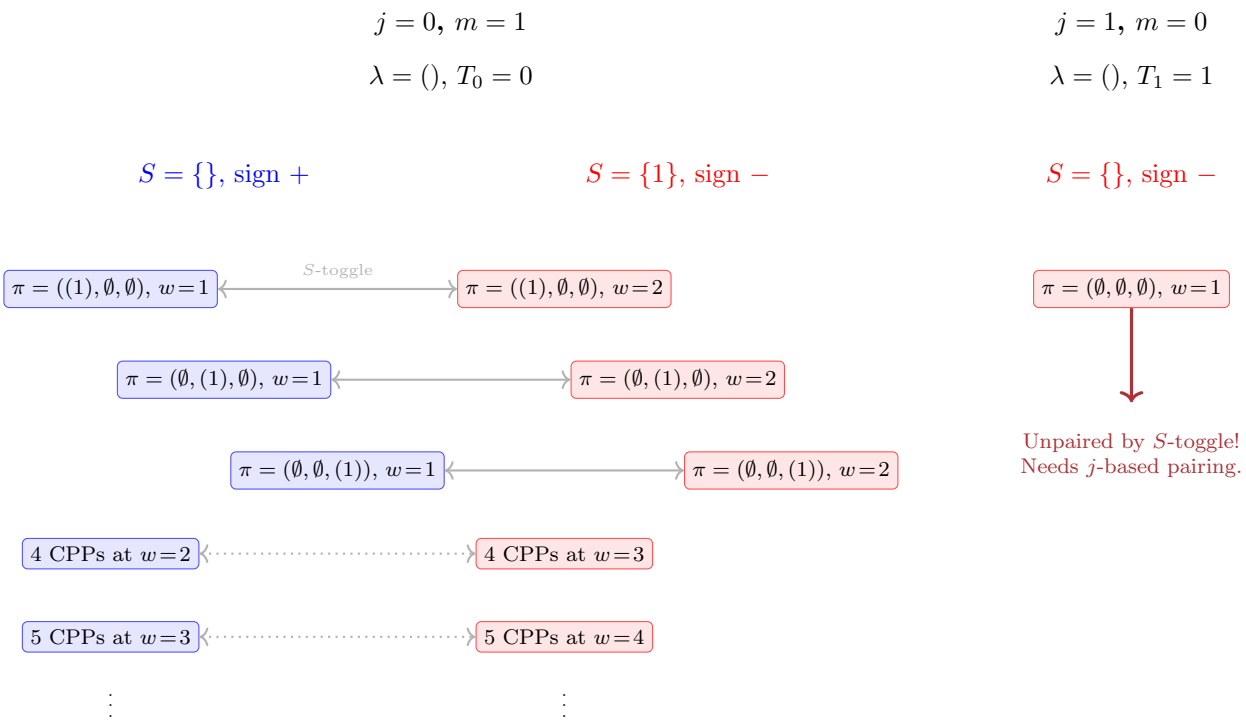
Net by weight

w	+	−	net	$[q^w]Q$
2	3	3	0	0
3	4+1	4	1	1
4	5	2	3	3
5	3	1	2	2
6	3	0	3	3
7	2	0	2	2
8	2	0	2	2
9	1	0	1	1
10	1	0	1	1
12	1	0	1	1

$Q_2(1) = 16 = 4^2$

Figure 6: Finest signed objects (j, λ, S, π) for $n = 1$, $c = (2, 1, 1)$

Expanding $(q)_1 = 1 - q$ gives subsets $S \subseteq \{1\}$. Sign $= (-1)^{j+|S|}$.



Key: The S-toggle involution pairs $(j{=}0, S{=}\{\}, \pi)$ with $(j{=}0, S{=}\{1\}, \pi)$, absorbing the $(1{-}q)$ factor that converts $g_1 \to h_1$. This is exactly the *injection* in disguise! But the $j{=}1$ negative object (empty CPP at weight 1) remains unpaired — it must cancel against one of the three $j{=}0$ weight-1 fixed points via a different mechanism.